50 FLOS for Hegra Conservation Project: QGIS, PostGIS and collaborative offline workflows

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This paper examines the use of FLOS software to record and manage conservation data for the monumental tombs at the UNESCO site of Hegra, Saudi Arabia. As part of the Hegra Conservation Project, led by Estia Restauri under the auspices of the Royal Commission for AlUla (RCU), ES Progetti e Sistemi was responsible for managing and archiving spatial data and descriptive records.

The system supports detailed condition mapping and georeferenced recording of constituent materials, deterioration patterns, structural damage and conservation interventions. Standardised recording procedures link mapped features to inspection records, providing a shared framework for documenting the tombs and their facades.

Built on QGIS and PostgreSQL/PostGIS, the platform enables conservators, archaeologists and other specialists to work collaboratively in the field, including where internet access is unreliable or unavailable. ES developed custom QGIS forms and plugins for offline editing and on-demand commits. Users collect and update data locally, then initiate synchronisation with the central geodatabase through the plugin. Each commit triggers checks on data structure and content, together with checks for conflicts arising from concurrent edits.

This workflow combines independent fieldwork with centralised data management, supporting consistent recording, data quality and coordination across documentation and conservation tasks. Automated export to ESRI File Geodatabase supports data exchange with other GIS environments.

The case illustrates how an established FLOS software stack, extended with project-specific forms and plugins, can meet the practical demands of collaborative heritage conservation. It highlights the role of customisation in connecting field recording, data validation and interoperability within a working professional system.

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